LEAP FINAL PROJECT PORTFOLIO • SCENARIO 1: DIABETES PROGRESSION

Analytical Report: Statistical Determinants of Diabetes Progression

Author: Data Analytics & Statistics Team | **Cohort:** 442 Patients | **Methodology:** OLS Linear Modeling

Executive Overview & Research Focus

This investigation evaluates baseline physiological and metabolic markers associated with 1-year diabetes disease progression among 442 adult patients. By following a structured statistical workflow, this report examines data integrity, details exploratory data distributions, examines pairwise linear dependencies, estimates simple and multiple ordinary least squares (OLS) regression models, verifies diagnostic validity, and outlines critical boundaries regarding observational inference.

> **Core Research Question:** Which baseline patient characteristics are most strongly associated with one-year quantitative diabetes progression, how do multi-variable regression models enhance explanatory capability, and what conclusions remain statistically defensible?

Question 1: Data Understanding, Integrity & Responsible Governance

The dataset consists of 442 rows, where each single row denotes an individual patient assessed at baseline and re-evaluated after one year. The primary outcome variable is **progression** (denoted as Y in the raw data), which reflects a continuous quantitative index of disease progression. Predictors encompass demographic indicators (`age`), physical measures (`bmi`, bp), a categorical variable for biological sex (`sex`), and six serum laboratory measurements (`s1` through `s6`).

Variable Name	Variable Role	Statistical Nature	Clinical Description
progression (Y)	Outcome	Continuous Numeric	Quantitative measure of disease progression 1 year post-baseline
age	Predictor	Continuous Numeric	Patient age in years (Range: 19.0 to 79.0)
sex	Predictor	Categorical Factor	Biologically coded (1, 2); factored to prevent continuous distortion
bmi	Predictor	Continuous Numeric	Body Mass Index (Weight in kg / Height in m^2)
bp	Predictor	Continuous Numeric	Mean arterial blood pressure (mmHg)
s1 – s6	Predictors	Continuous Numeric	Blood serum tests: Total Chol (s1), LDL (s2), HDL (s3), Ratio (s4), Log Triglycerides (s5), Glucose (s6)

Data Hygiene & Ethical Considerations

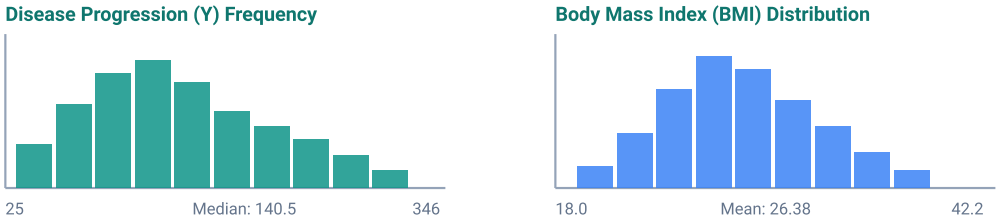
Inspection in R via `colSums(is.na(diabetes))` and `duplicated()` revealed exactly **0 missing values** and **0 duplicate rows** across all 442 records. Biological metrics are bounded within realistic ranges. Ethical data governance dictates strict patient anonymization and cautious interpretation: because healthcare data informs clinical interventions, non-causal associations must never be communicated as validated prescriptive targets.

Question 2: Exploratory Data Analysis & Empirical Distributions

Exploratory evaluation establishes the structural properties, central tendency, and dispersion of clinical variables prior to statistical fitting. Descriptive analysis reveals that the primary outcome, `progression`, spans from 25.0 to 346.0 units, demonstrating slight right-skewness while preserving an unbroken central density.

Metric Name	Mean (μ)	Std. Dev. (σ)	Median	IQR	Min – Max	Distributional Form
progression	152.13	77.09	140.50	124.50	25.0 – 346.0	Mild right-skew; unimodal
bmi	26.38	4.42	25.70	6.08	18.0 – 42.2	Near symmetric; slight right tail
bp	94.65	13.83	93.00	21.00	62.0 – 133.0	Symmetric bell curve
age	48.52	13.11	50.00	20.75	19.0 – 79.0	Symmetric adult distribution
s5 (log trig)	4.64	0.52	4.62	0.72	3.26 – 6.11	Standard Gaussian profile

Comparative Distribution Profile: Progression vs. BMI



Analytical Takeaways Before Modeling

Both `progression` and `bmi` show no detached outlier clusters. The median BMI of 25.70 indicates an overweight cohort, which aligns with clinical expectations for adult type 2 diabetes research. Standard deviation values remain stable, confirming that data scaling or non-linear root transforms are not strictly required for initial linear baseline modeling.

Question 3: Pairwise Bivariate Relationships & Collinearity Checks

Pearson correlation coefficients were computed across all continuous metrics (excluding categorical `sex`). The correlation matrix identifies two dominant predictors positively linked with disease progression: **BMI** (r = 0.586) and **s5** (r = 0.566). Arterial blood pressure (**bp**) also exhibits a solid correlation (r = 0.441). High-density lipoprotein (**s3**) yields a moderate negative relationship (r = −0.395), demonstrating the cardioprotective biological role of HDL.

Predictor	Pearson r with Progression	Primary Cross-Correlations	Collinearity Risk Assessment
bmi	+0.586	r(bp) = 0.395, r(s5) = 0.446	Low collinearity; high unique predictive variance
s5 (log trig)	+0.566	r(s4) = 0.618, r(s1) = 0.515	Moderate inter-correlation; strong signal strength
bp	+0.441	r(bmi) = 0.395, r(s5) = 0.398	Low collinearity; reliable hemodynamic proxy
s4 (ratio)	+0.430	r(s3) = −0.738, r(s5) = 0.618	High redundancy with s3 and s5; potential instability
s3 (HDL)	−0.395	r(s4) = −0.738, r(s2) = −0.351	Negative protective association; redundant with s4
s1 (Total Chol)	+0.212	r(s2) = +0.897	Extreme collinearity with LDL (s2); do not co-enter

> **Multicollinearity Risk Identified:** The correlation between total serum cholesterol (`s1`) and LDL cholesterol (`s2`) is **r = 0.897**. Introducing both variables simultaneously would artificially inflate standard errors and destabilize parameter sign attribution. Consequently, serum dimensions must be selectively entered into multiple regression models.

Variable Selection Justification for Multiple Regression

Based on the bivariate matrix, the candidate trio selected for multi-variable regression is `bmi + bp + s5`. This combination captures three distinct pathophysiological mechanisms: adiposity/metabolic load (BMI), systemic vascular pressure (BP), and circulating lipid dysregulation (s5). Crucially, their shared correlations are moderate (0.395 to 0.446), providing complementary statistical information while avoiding multicollinearity.

Question 4: Model Estimation & Comparative Regression Analysis

Two sequential OLS specifications were estimated: a simple bivariate baseline model using `bmi`, followed by the tri-variable model incorporating `bmi`, `bp`, and `s5`. Parameter estimates, standard errors, test statistics, and 95% confidence intervals are summarized below.

Model Specification	Parameter Term	Estimate (β)	Std. Error	t-statistic	p-value	95% Confidence Interval
Simple Model	Intercept	−117.77	17.98	−6.55	< 0.0001	[−153.11, −82.43]
(R^2 = 0.344)	**bmi**	+10.23	0.67	15.20	< 0.0001	[+8.91, +11.56]
Multiple Model	Intercept	−334.88	26.96	−12.42	< 0.0001	[−387.87, −281.89]
(R^2 = 0.480)	**bmi**	+6.50	0.70	9.32	< 0.0001	[+5.13, +7.87]
(Adj R^2 = 0.477)	**bp**	+0.90	0.22	4.17	< 0.0001	[+0.48, +1.33]
(σ = 55.77)	**s5**	+49.58	5.89	8.42	< 0.0001	[+38.00, +61.15]

Interpretation of Coefficients & Model Comparison

In the simple regression, each 1-unit increase in BMI corresponds to a 10.23-point increase in expected progression. In the multiple model, after *holding blood pressure and s5 constant*, the estimated BMI slope attenuates to 6.50 units. This shows that a portion of the apparent bivariate association between BMI and progression was shared with elevated blood pressure and serum lipid levels.

Evaluation Metric	Simple Model (BMI)	Multiple Model (BMI + BP + S5)	Model Progression Difference
Explained Variance (R^2)	0.3439 (34.4%)	0.4803 (48.0%)	+13.64 percentage points
Adjusted R^2	0.3424 (34.2%)	0.4767 (47.7%)	+13.43 percentage points
Residual Standard Error (σ)	62.51 units	55.77 units	−6.74 units (Improved accuracy)

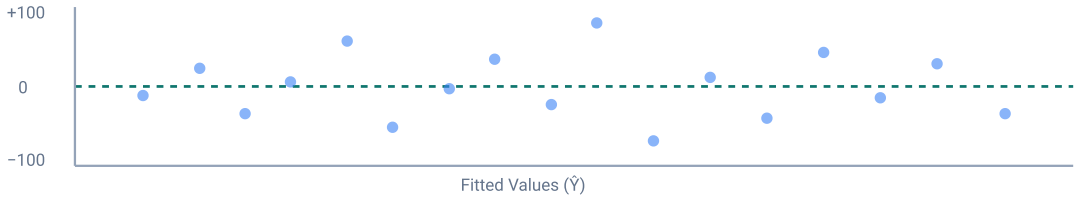
Comparative Conclusion

The multiple model improves explanatory power over the simple model: the Adjusted R^2 increases from 0.342 to 0.477, and the residual standard error drops by 6.74 units. However, high R^2 alone does not establish complete clinical adequacy, as unexplained variation remains substantial.

Question 5: Model Evaluation, Diagnostics & Uncertainty

Evaluation of the multiple regression requires assessing residual patterns and formal collinearity checks. Variance Inflation Factors (VIF) yield **1.34 for BMI**, **1.27 for BP**, and **1.34 for S5**. Because all VIF figures sit well below the conservative rule-of-thumb cutoff of 5.0, parameter stability is confirmed.

Diagnostic Plot: Residuals vs. Fitted Values (Multiple Model)



Residuals distribute symmetrically above and below the zero line without strong non-linear curvature, indicating satisfactory homoskedasticity across the range of fitted values.

Question 6: Causal Boundaries, Confounding & Conclusions

The diabetes dataset is derived from an **observational cohort study**, not a randomized controlled clinical trial. Consequently, **regression coefficients reflect statistical associations rather than direct causal effects**. Inferring that therapeutically reducing a patient's BMI will automatically reduce their disease progression score by 6.50 points confuses statistical conditioning with clinical intervention.

> **Key Limitations & Omitted Variable Bias:** The dataset lacks tracking of pharmacological therapies (e.g., insulin or metformin), dietary adherence, exercise routines, smoking status, and socioeconomic background. Furthermore, 52% of outcome variance remains unaccounted for (σ = 55.77).

Defensible Conclusions vs. Unwarranted Inferences

Supported Conclusions: Baseline BMI, blood pressure, and serum triglycerides (s5) serve as robust prognostic risk markers for identifying patients at elevated risk of rapid one-year disease progression.
Unsupported Inferences: Claiming causal efficacy for specific biomarker interventions or extrapolating these predictions to pediatric, geriatric, or racially diverse patient groups without prior external validation.